PostgreSQL 16 Advanced Indexing & EXPLAIN Plans

PostgreSQL 16.6
VERIFIED RUNTIME

KEY SYNTAX & IDIOMS

Explain Analyze Plan
```
EXPLAIN (ANALYZE, BUFFERS) SELECT ...;
```
Inspects actual execution runtime costs

B-Tree Index Creation
```
CREATE INDEX idx_users_email ON users(email);
```
Accelerates lookups and range scans

Common Table Expression
```
WITH ranked AS (SELECT ...) SELECT * FROM ranked;
```
Modularizes complex query steps

Safe Upsert ON CONFLICT
```
INSERT INTO ... ON CONFLICT (id) DO UPDATE ...;
```
Atomic insert or update operation

COMMON PITFALLS & FIXES

Pitfall: DROP DATABASE in application code
Rule: Strictly block destructive DDL in verification sandboxes

Pitfall: SELECT * in high-volume tables
Rule: Specify explicit column projections to save buffer reads

Pitfall: Sequential scans on foreign keys
Rule: Always index foreign key columns used in joins

TINY VERIFIED EXAMPLE: POSTGRESQL 16 CTE QUERY

SOURCE CODE:
```
WITH demo AS (SELECT 'Verified' AS status)
SELECT status FROM demo;
```

REAL CAPTURED OUTPUT:
```
status: Verified
```